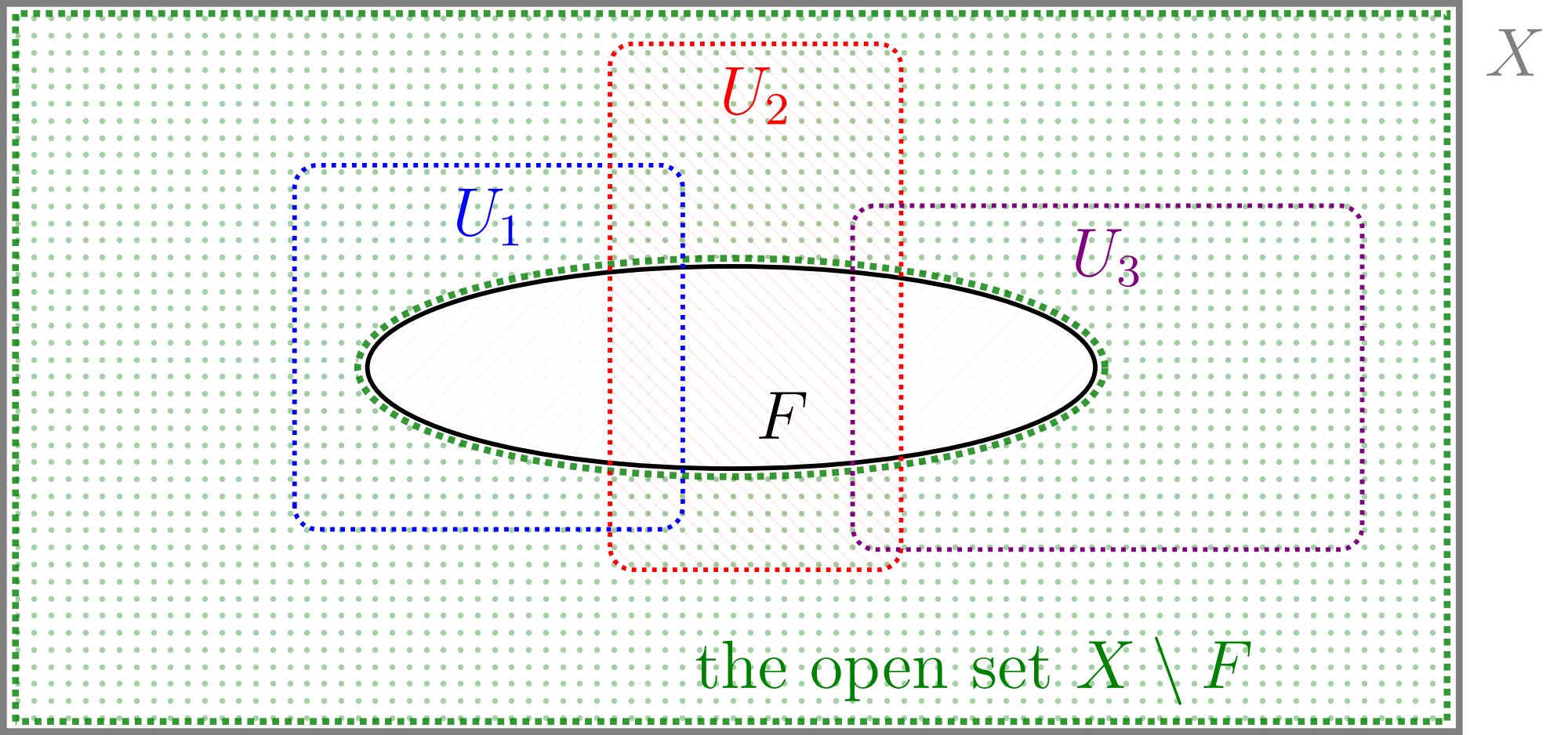
X
U_2
U_1
U_3
F
the open set $X \setminus F$